9) Ridge Regression

```
import numpy as np
from sklearn.datasets import
load_diabetes
from sklearn.linear_model import Ridge
from sklearn.model_selection import
train_test_split
from sklearn.metrics import
mean_squared_error, r2_score
# Load dataset
data = load_diabetes()
X = data.data
Y = data.target
# Split data
X_train, X_test, Y_train, Y_test =
    train_test_split (X, Y, test_size= 0.2, random_state=42)
# Initialize Ridge regression (alpha=1.0)
ridge = Ridge (alpha=1.0)
ridge.fit(X_train, Y_train)
# prediction
Y_pred = ridge.predict (X_test)
# Evaluation
print ("simple Ridge Regression")
print ("MSE:", mean_squared_error (Y_test, Y_pred))
print ("R² score:", r2_score (Y_test, Y_pred))
print ("coefficients (first 5):", ridge.coef_[:5])
print ("Intercept:", ridge.intercept_)
```

OUTPUT

```
Simple Ridge Regression
MSE: 3077.4159388 2723
R² score: 0.41915292635986545
Coefficients (first 5): [45.36737726 -76.66608563
    291.33883165 198.99581745 -0.53030959]
Intercept: 152.24167521111 3
```